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Rotatable heat exchanger for a gas turbine engine

Abstract

A rotatable heat exchanger includes a first manifold assembly, a second manifold assembly, and a core extending axially from the first manifold assembly to the second manifold assembly. The first manifold assembly, the core, and the second manifold assembly are formed to mount around and rotate with a shaft. The core includes a plurality of helical passages that extend from the first manifold assembly to the second manifold assembly. The plurality of helical passages includes a plurality of first-fluid passages fluidly coupled to the first manifold assembly and to the second manifold assembly, and a plurality of second-fluid passages fluidly coupled to the first manifold assembly and the second manifold assembly.

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Background/Summary

FIELD

(1) The present disclosure relates to a gas turbine engine, and more particularly to a rotatable heat exchanger for a gas turbine engine.

BACKGROUND

(2) Gas turbine engines, such as turbofan engines, may be used for aircraft propulsion. A turbofan engine generally includes a bypass fan section and a turbomachine such as a gas turbine engine to drive the bypass fan. The turbomachine generally includes a compressor section, a combustion

section, and a turbine section in a serial flow arrangement. Both the compressor section and the turbine section are driven by a rotor shaft.

(3) A bearing compartment is used to house a bearing assembly which enables smooth rotation of the rotor shaft in gas turbine engines. The bearing compartment houses a bearing cavity and buffer cavity which are partly separated by seals (e.g. carbon/lift-off seals). The bearing cavity is usually supplied with lubrication oil to reduce friction and the seals to keep the oil in-place.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A full and enabling disclosure of the present disclosure, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:

(2) FIG. 1 is a cross-sectional view of a gas turbine engine in accordance with an exemplary aspect of the present disclosure.

(3) FIG. 2 is a schematic view of an exemplary bearing compartment of a gas turbine engine according to exemplary embodiments of the present disclosure.

(4) FIG. 3 is a cross-sectional schematic view of an exemplary embodiment of a rotatable heat exchanger according to exemplary embodiments of the present disclosure.

(5) FIG. 4 is a schematic view of an exemplary embodiment of the rotatable heat exchanger of FIG. 3.

(6) FIG. 5 is an enlarged cross-sectional schematic view of a first end portion of the rotatable heat exchanger including a first manifold assembly as shown in FIG. 3, according to exemplary embodiments of the present disclosure.

(7) FIG. 6 is an enlarged cross-sectional schematic view of a second end portion of the rotatable heat exchanger including a first manifold assembly as shown in FIG. 3, according to exemplary embodiments of the present disclosure.

(8) FIG. 7 is a perspective view of a rotatable heat exchanger as shown in FIG. 3, with an outer casing removed for clarity, according to an exemplary embodiment of the present disclosure.

(9) FIG. 8 is an enlarged view of a portion of a first end of a core of the rotatable heat exchanger with an outer casing removed and a first manifold assembly attached, according to exemplary embodiments of the present disclosure.

(10) FIG. 9 is an enlarged view of a portion of a second end of a core of the rotatable heat exchanger with an outer casing removed and a second manifold assembly attached, according to exemplary embodiments of the present disclosure.

(11) FIG. 10 is a schematic of a bearing cooling system according to an exemplary embodiment of the present disclosure.

(12) FIG. 11 is a cross-sectional schematic view of an alternate embodiment of a rotatable heat exchanger, according to exemplary embodiments of the present disclosure.

(13) FIG. 12 is a schematic of a bearing cooling system according to an exemplary embodiment of the present disclosure.

(14) FIG. 13 provides a table including numerical values corresponding to several rotatable heat exchangers falling within the ranges depicted in FIG. 14.

(15) FIG. 14 is a plot of RHECP on a Y-axis and shaft speed on an X-axis.

DETAILED DESCRIPTION

(16) Reference will now be made in detail to present embodiments of the disclosure, one or more examples of which are illustrated in the accompanying drawings. The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and description have been used to refer to like or similar parts of the disclosure.

(17) The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations. Additionally, unless specifically identified otherwise, all embodiments described herein should be considered exemplary. The singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise. The term “at least one of” in the context of, e.g., “at least one of A, B, and C” refers to only A, only B, only C, or any combination of A, B, and C.

(18) As used herein, the terms “first”, “second”, and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components. Furthermore, the terms “upstream” and “downstream” refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows.

(19) The term “turbomachine” refers to a machine including one or more compressors, a heat generating section (e.g., a combustion section), and one or more turbines that together generate a torque output. The term “gas turbine engine” refers to an engine having a turbomachine as all or a portion of its power source. Example gas turbine engines include turbofan engines, turboprop engines, turbojet engines, turboshaft engines, etc., as well as hybrid-electric versions of one or more of these engines.

(20) As used herein, the terms “additively manufactured” or “additive manufacturing techniques or processes” refer generally to manufacturing processes wherein successive layers of material(s) are provided on each other to “build-up,” layer-by-layer, a three-dimensional component. The successive layers generally fuse together to form a monolithic component which may have a variety of integral sub-components. Although additive manufacturing technology is described herein as enabling fabrication of complex objects by building objects point-by-point, layer-by-layer, typically in a vertical direction, other methods of fabrication are possible and within the scope of the present subject matter. For example, although the discussion herein refers to the addition of material to form successive layers, one skilled in the art will appreciate that the methods and structures disclosed herein may be practiced with any additive manufacturing technique or manufacturing technology. For example, embodiments of the present disclosure may use layer-additive processes, layer-subtractive processes, or hybrid processes.

(21) The terms “coupled,” “fixed,” “attached to,” and the like refer to both direct coupling, fixing, or attaching, as well as indirect coupling, fixing, or attaching through one or more intermediate components or features, unless otherwise specified herein.

(22) The phrases “from X to Y” and “between X and Y” each refers to a range of values inclusive of the endpoints (i.e., refers to a range of values that includes both X and Y).

(23) The term “100% shaft speed operating condition” refers to an operating condition of a gas turbine engine wherein at least one shaft of the gas turbine engine is operated at rotational speed corresponding to its 100% shaft speed capacity. The 100% shaft speed capacity is a shaft operating condition that may occur during a takeoff operating condition, a maximum power cruise operating condition, or other high power operating condition. In some configurations, the 100% shaft speed operating condition may be less than a redline operating speed for the shaft.

(24) Generally, a turbofan engine includes a fan and a turbomachine, with the turbomachine rotating the fan to generate thrust. The turbomachine includes a compressor section, a combustion section, a turbine section, and an exhaust section and defines a working gas flowpath therethrough. Compressed air from the compressor section (i.e., bleed air) is utilized for a variety of functions within the turbomachine, such as to pressurize bearing cavities, cool components of the turbomachine, etc. Depending on where the bleed air is taken from the compressor section, the bleed air may be at a relatively high temperature, such that it needs to be cooled to accomplish certain functions.

(25) In particular, the inventor of the present disclosure sought out to provide bleed air from an HP compressor of the turbomachine to a sump cavity to pressurize the sump cavity. The inventor found, however, that when the bleed air is provided at a relatively high temperature, as is typical of the bleed air from a stage of the HP compressor having airflow at a sufficiently high pressure to meet the pressurization needs for the sump cavity, the bleed air degrades the seals positioned within the sump cavity designed to prevent oil leakage. Therefore, the inventor recognized a need to cool the bleed air upstream of the bleed cavity to avoid this issue.

(26) The inventor recognized that a rotatable heat exchanger can be coupled to a shaft of the turbomachine to cool the bleed air. The rotatable heat exchanger transfers heat from the bleed air to a heat exchange fluid, such as, e.g., lubrication oil, fuel, etc. Previous thinking has led to the conclusion that including such a heat exchanger would create excessive drag on the driving shaft and would be too difficult to control (or too difficult to control flow through such a heat exchanger) to justify any thermal management benefits associated with inclusion within the turbomachine.

(27) However, the inventor discovered, unexpectedly, in the course of designing a rotatable heat exchanger for the above noted need that the costs associated with inclusion of the rotatable heat exchanger on the shaft of the turbomachine may be overcome by the thermal management benefits, contrary to previous thinking and expectations. In particular, the inventor discovered during the course of designing several rotatable heat exchangers to be used with several engine architectures of varying thrust classes and mission requirements (including the rotatable heat exchangers and engines illustrated and described in detail herein), a relationship exists among an amount of airflow available to the heat exchanger (as may be indicated by bypass ratio and fan radius), various geometric ratios of the passages through the heat exchanger, and an expected rotational speed of the rotatable heat exchanger, whereby including a rotatable heat exchanger in accordance with one or more of the exemplary aspects described herein results in a net benefit to the overall engine design. As noted, previous thinking was that the cost for including a rotatable heat exchanger coupled to a shaft of the turbomachine to reduce a temperature of a bleed airflow was too prohibitive, as compared to the benefits of doing so.

(28) With a goal of arriving at an improved gas turbine engine capable of providing cooled bleed air to the buffer cavity(ies), the inventor proceeded in the manner of designing gas turbine engines having a rotatable heat exchanger coupled to a shaft of a turbomachine with various amounts of airflow available to the rotatable heat exchanger, with a variety of geometric ratios of the passages through the heat exchanger, and operable at various rotational speeds; checking operability and thermal rejection characteristics of the designed gas turbine engine; redesigning the gas turbine engine to vary the noted parameters based on the impact on other aspects of the gas turbine engine; rechecking the operability and thermal rejection characteristics of the redesigned gas turbine engine; etc. during the design of several different types of rotatable heat exchangers and gas turbine engines, including the rotatable heat exchangers and gas turbine engines described below with reference to FIGS. 1 through 14, which are described below in greater detail.

(29) For example, one such rotatable heat exchanger of the present disclosure discovered by the inventor may be used to cool a bleed air provided to pressurize a bearing compartment. The bearing compartment houses a bearing assembly, which enables smooth rotation of a low-pressure (LP) rotor shaft in the gas turbine engine. The bearing compartment also houses a bearing cavity and a buffer cavity which are partly separated by seals (e.g., carbon/lift-off seals). The bearing cavity is usually supplied with lubrication oil to reduce friction. Seals are used to keep the oil in-place. The seals are engaged to seal runners that are secured to the LP shaft.

(30) To ensure zero or minimal oil leakage, high pressure compressed air from a compressor section of the gas turbine engine is supplied to the buffer cavity to create an air gap at a higher pressure relative to that of the lubricating oil in the bearing cavity. However, since the compressed air is bled-off from the compressor stage, it is typically supplied at high temperature, which can cause reduced lifespan of the seals and other components downstream, possibly resulting in oil

leakage from the bearing cavity. This disclosure introduces a compact shaft-mounted rotatable heat exchanger for cooling the compressed air, and subsequently the seals, using a fluid such as the lubricating oil as the cooling fluid. The rotatable heat exchanger may be designed to achieve the above noted goals, while minimizing the costs associated with inclusion of the rotatable heat exchanger.

(31) In one embodiment, the rotatable heat exchanger includes a core having multiple cylindrical plates with different diameters that surround the rotor shaft and are concentric about the shaft's rotational axis. The concentric cylindrical plates are linked with each other and connected to inner and outer casings by radially spaced and circumferentially-arranged helical plates with one or more turns, which also provides structural support for the core and acting as channel walls. This forms helical flow channels with one or more turns spanning across the axial ends of the cylindrical plates. The axial ends of the core are connected to two manifold assemblies that each allow inflow and/or outflow of two different flow streams, thereby forming a two-fluid rotatable heat exchanger that is then linked to the seal runners.

(32) Referring now to the drawings, wherein identical numerals indicate the same elements throughout the figures, FIG. 1 is a schematic cross-sectional view of a gas turbine engine in accordance with an exemplary embodiment of the present disclosure. More particularly, for the embodiment of FIG. 1, the gas turbine engine is configured as a high-bypass turbofan jet engine 10, referred to herein as “turbofan engine 10.” As shown in FIG. 1, the turbofan engine 10 defines an axial direction “A” (extending parallel to a longitudinal centerline or axis 12 provided for reference), a radial direction “R”, and a circumferential direction “C” extending about the axial direction A. In general, the turbofan engine 10 includes a fan section 14 and a turbomachine 16 disposed downstream from the fan section 14.

(33) The exemplary turbomachine 16 depicted generally includes a substantially tubular outer casing 18 that defines an annular inlet 20. The outer casing 18 encases and the turbomachine 16 includes, in serial flow relationship, a compressor section 21 including a booster or low pressure (LP) compressor 22 and a high pressure (HP) compressor 24; a combustion section 26; a turbine section including a high pressure (HP) turbine 28 and a low pressure (LP) turbine 30; and a jet exhaust nozzle section 32. A high pressure (HP) shaft or spool 34 drivingly connects the HP turbine 28 to the HP compressor 24. A low pressure (LP) shaft or spool 36 drivingly connects the LP turbine 30 to the LP compressor 22. Accordingly, the LP shaft 36 and HP shaft 34 are each rotary components, rotating about the axial direction A during operation of the turbofan engine 10.

(34) Referring still to the embodiment of FIG. 1, the fan section 14 includes a variable pitch fan 38 having a plurality of fan blades 40 coupled to a disk 42 in a spaced apart manner. As depicted, the fan blades 40 extend outwardly from disk 42 generally along the radial direction R and collectively define a fan radius (RF) equal to a distance from the longitudinal axis 12 to the radially outer tips of the fan blades 40. Each fan blade 40 is rotatable relative to the disk 42 about a pitch axis P by virtue of the fan blades 40 being operatively coupled to a suitable pitch change mechanism 44 configured to collectively vary the pitch of the fan blades 40 in unison. The fan blades 40, disk 42, and pitch change mechanism 44 are together rotatable about the longitudinal axis 12 by LP shaft 36 across a power gear box 46. The power gear box 46 includes a plurality of gears for adjusting the rotational speed of the fan 38 relative to the LP shaft 36 to a more efficient rotational fan speed. More particularly, the fan section includes a fan shaft rotatable by the LP shaft 36 across the power gearbox 46. Accordingly, the fan shaft may also be considered a rotary component, and is similarly supported by one or more bearings.

(35) Referring still to the exemplary embodiment of FIG. 1, the disk 42 is covered by a rotatable front hub 48 aerodynamically contoured to promote an airflow through the plurality of fan blades 40. The front hub 48 defines a hub radius (RH) equal to a distance from the longitudinal axis 12 to a radially outermost point of the front hub 48. Additionally, the exemplary fan section 14 includes an annular fan casing or nacelle 50 that circumferentially surrounds the fan 38 and/or at least a

portion of the turbomachine **16**. The exemplary nacelle **50** is supported relative to the turbomachine **16** by a plurality of circumferentially-spaced outlet guide vanes **52**. Moreover, a downstream section **54** of the nacelle **50** extends over an outer portion of the turbomachine **16** so as to define a bypass airflow passage **56** therebetween.

(36) During operation of the turbofan engine **10**, a volume of air **58** enters the turbofan engine **10** through an associated inlet **60** of the nacelle **50** and/or fan section **14**. As the volume of air **58** passes across the fan blades **40**, a first portion of the air **58** as indicated by arrows **62** is directed or routed into the bypass airflow passage **56** and a second portion of the air **58** as indicated by arrow **64** is directed or routed into the core air flowpath **37**, or more specifically into the LP compressor **22**. The ratio between the first portion of air **62** and the second portion of air **64** is commonly known as a bypass ratio. The pressure of the second portion of air **64** is then increased as it is routed through the high pressure (HP) compressor **24** and into the combustion section **26**, where it is mixed with fuel and burned to provide combustion gases **66**.

(37) The combustion gases **66** are routed through the HP turbine **28** where a portion of thermal and/or kinetic energy from the combustion gases **66** is extracted via sequential stages of HP turbine stator vanes **68** that are coupled to the outer casing **18** and HP turbine rotor blades **70** that are coupled to the HP shaft **34**, thus causing the HP shaft **34** to rotate, thereby supporting operation of the HP compressor **24**. The combustion gases **66** are then routed through the LP turbine **30** where a second portion of thermal and kinetic energy is extracted from the combustion gases **66** via sequential stages of LP turbine stator vanes **72** that are coupled to the outer casing **18** and LP turbine rotor blades **74** that are coupled to the LP shaft **36**, thus causing the LP shaft **36** to rotate, thereby supporting operation of the LP compressor **22** and/or rotation of the fan **38**.

(38) The combustion gases **66** are subsequently routed through the jet exhaust nozzle section **32** of the turbomachine **16** to provide propulsive thrust. Simultaneously, the pressure of the first portion of air **62** is substantially increased as the first portion of air **62** is routed through the bypass airflow passage **56** before it is exhausted from a fan nozzle exhaust section **76** of the turbofan engine **10**, also providing propulsive thrust. The HP turbine **28**, the LP turbine **30**, and the jet exhaust nozzle section **32** at least partially define a hot gas path **78** for routing the combustion gases **66** through the turbomachine **16**.

(39) It should be appreciated, however, that the exemplary turbofan engine **10** depicted in FIG. **1** is provided by way of example only, and that in other exemplary embodiments, the turbofan engine **10** may have any other suitable configuration. It should also be appreciated, that in still other exemplary embodiments, aspects of the present disclosure may be incorporated into any other suitable gas turbine engine. For example, in other exemplary embodiments, aspects of the present disclosure may be incorporated into, e.g., a turboprop engine, a turboshaft engine, or a turbojet engine. Further, in still other embodiments, aspects of the present disclosure may be incorporated into any other suitable turbomachine, including, without limitation, a power generation gas turbine engine, a steam turbine, a centrifugal compressor, and/or a turbocharger.

(40) In order to support rotary components, such as the fan section **14**, the LP shaft **36** and the HP shaft **34**, the turbofan engine **10** includes a plurality of bearing assemblies **80** attached to various structural components within the turbofan engine **10**. Specifically, for the embodiment depicted the bearing assemblies **80** facilitate rotation of, e.g., the LP shaft **36** and HP shaft **34** and dampen vibrational energy imparted to bearing assemblies **80** during operation of the turbofan engine **10**. Although the bearing assemblies **80** are described and illustrated as being located generally at forward and aft ends of the respective LP shaft **36** and HP shaft **34**, the bearing assemblies **80** may additionally, or alternatively, be located at any specifically pre-determined location along the LP shaft **36** and HP shaft **34** including, but not limited to, central or mid-span regions of the HP and LP shafts **34**, **36**, or other locations along HP and LP shafts **34**, **36** where the use of conventional bearing assemblies **80** would present significant design challenges. The exemplary bearing assemblies **80** may include radial support bearings for supporting static and dynamic forces along

the radial direction R, as well as axial support bearings, or thrust bearings, for supporting static and dynamic forces along the axial direction A. Further, in exemplary embodiments, the bearing assemblies **80** are oil-lubricated bearings. For example, in one embodiment, conventional oil-lubricated bearings may be located at the ends of HP and LP shafts **34**, **36**, and one or more bearing assemblies **80** may be located along central or mid-span regions of HP and LP shafts **34**, **36**.

(41) FIG. 2 is a schematic view of an exemplary bearing compartment **82** of the turbofan engine **10** according to various embodiments of the present disclosure. In exemplary embodiments, the bearing compartment **82** is used to house one or more respective bearing assemblies **80** which enables smooth rotation of the LP shaft **36** of the turbofan engine **10** (FIG. 1). As shown in FIG. 2, the bearing compartment **82** also houses a bearing cavity **84** and a buffer cavity **86**, which are partly separated by seals **88** (only one shown), e.g., carbon/lift-off seals. The bearing cavity **84** may be supplied with a lubricant or lubricating oil **90** to reduce friction. The seals **88** are provided to keep the lubricating oil **90** from leaking from the bearing cavity in operation. The seals **88** are engaged to seal runners (not shown) that are secured to the LP shaft **36** (FIG. 1).

(42) To ensure minimum or zero oil leakage around the seals **88**, high pressure compressed air **92** from one or both the LP compressor **22** and the HP compressor **24** of the compressor section **21** is supplied to the buffer cavity **86** to create an air gap **94** at a higher pressure, relative to that of the lubricating oil **90**, in the bearing cavity **84**. However, since the compressed air **92** is bled-off from the compressor section **21**, it is typically supplied at a relatively high temperature, which can reduce lifespan of the seals **88** and other components downstream, potentially resulting in oil leakage.

(43) Referring to FIGS. 1 and 2 collectively, in exemplary embodiments of the present disclosure, the turbofan engine **10** includes a compact shaft-mounted rotatable heat exchanger **200** hereinafter referred to as “rotatable heat exchanger **200**”. In operation, the rotatable heat exchanger **200** cools the compressed air **92** fed to the buffer cavity **86** and subsequently to the seals **88**, using the lubricating oil **90** as a cooling fluid.

(44) FIG. 3 provides a cross-sectioned schematic view of an exemplary embodiment of the rotatable heat exchanger **200** according to exemplary embodiments of the present disclosure. FIG. 4 provides a schematic view of the rotatable heat exchanger **200** of FIG. 3.

(45) As shown in FIGS. 3 and 4, the rotatable heat exchanger **200** includes an outer casing **202** radially spaced from and concentrically aligned with an annular inner casing **204**. The annular inner casing **204** and the outer casing **202** at least partially define a core **206** of the rotatable heat exchanger **200**. The core **206**, and more specifically for the embodiment shown, the outer casing **202**, defines a heat exchanger radius (R.sub.HX) of the core **206** of the rotatable heat exchanger **200**.

(46) In exemplary embodiments, the rotatable heat exchanger **200** includes a first manifold assembly **208** at a first end **210** of the core **206**. A second manifold assembly **212** is disposed at a second end **214** of the core **206**. The first end **210** and the second end **214** of the core **206** are axially spaced apart with respect an axial centerline **216** of the rotatable heat exchanger **200** and/or the longitudinal axis **12** of the turbofan engine **10**, as shown in FIG. 1. The first manifold assembly **208** and/or the second manifold assembly **212** may be formed from multiple components assembled together or may be formed as singular bodies using additive manufacturing techniques known in the industry. Forming the components singularly via additive manufacturing provides for reduced weight, fewer connection joints resulting in less opportunities for leakage, etc.

(47) As discussed, the rotatable heat exchanger **200** is coupled to a shaft **201**. More specifically, the first manifold assembly **208**, the core **206** and the second manifold assembly **212** are annularly or ring shaped to fit around and be coupled to the shaft **201**, which may be one of the LP shaft **36** or the HP shaft **34**. The shaft **201** defines a shaft radius (R.sub.S). In such a manner, it will be appreciated that the rotatable heat exchanger **200** may be configured to rotate at a rotational speed (N.sub.S) in a circumferential direction C equal to a rotational speed of the shaft when the engine is

operated at a 100% shaft speed operating condition

(48) FIG. 5 provides an enlarged cross-sectional schematic view of a portion of the rotatable heat exchanger **200** including the first manifold assembly **208** as shown in FIG. 3, according to exemplary embodiments of the present disclosure. FIG. 6 provides an enlarged cross-sectional schematic view of a portion of the rotatable heat exchanger **200** including the second manifold assembly **212** as shown in FIG. 3, according to exemplary embodiments of the present disclosure.

(49) As shown in FIG. 5, the first manifold assembly **208** includes/defines a first-fluid inlet manifold **218**. In exemplary embodiments, the first-fluid inlet manifold **218** is annularly shaped. A first-fluid inlet **220** provides for fluid communication into the first-fluid inlet manifold **218**. The first manifold assembly **208** further includes a second-fluid outlet manifold **222**. In exemplary embodiments, the second-fluid outlet manifold **222** is annularly shaped. A second-fluid outlet **224** provides for fluid communication out of the second-fluid outlet manifold **222**.

(50) As shown in FIG. 6, the second manifold assembly **212** includes/defines a first-fluid outlet manifold **226**. In exemplary embodiments, the first-fluid outlet manifold **226** is annularly shaped. A first-fluid outlet **228** provides for fluid communication out of the first-fluid outlet manifold **226**. The second manifold assembly **212** further includes a second-fluid inlet manifold **230**. In exemplary embodiments, the second-fluid inlet manifold **230** is annularly shaped. A second-fluid inlet **232** provides for fluid communication into the second-fluid inlet manifold **230**. In certain embodiments the second-fluid inlet **232** may be formed from one or more apertures **234** defined along an inner wall **236** of the second manifold assembly **212** or the annular inner casing **204**. The one or more apertures **234** may be circumferentially spaced about the inner wall **236**.

(51) As shown in FIGS. 3-6 collectively, the core **206** includes a plurality of helical passages **238** concentrically formed about the annular inner casing **204** and/or axial centerline **216** and stacked radially outwardly from the annular inner casing **204**. As shown in FIGS. 5 and 6, each respective helical passage **238** of the plurality of helical passages **238** is formed between a respective pair of radially spaced walls or plates **240**. The radially spaced walls or plates **240** may be cylindrically or annularly shape about the annular inner casing **204** and/or the axial centerline **216**.

(52) Referring particularly to FIG. 4, the plurality of helical passages **238** includes a first passage **238A**. The first passage **238A** extends from an inlet **239-1** to an outlet **239-2** and defines a passage length (L.sub.P; not labeled) from the inlet **239-1** to the outlet **239-2**, which is an average passage length of the first passage **238A**. Additionally, as will be appreciated from the Callout A in FIG. 4, the first passage **238A** additionally defines a passage hydraulic diameter (D.sub.P), which is an average hydraulic diameter of the first passage **238A** along the passage length (L.sub.P). Briefly, it will further be appreciated that the first passage defines a head loss ($\Delta H_{\text{sub.channel}}$) across the first passage **238A**, which is an average head loss across the first passage **238A**, when the engine is operated at a 100% shaft speed operating condition. As used herein, the term “head loss” refers to a pressure lost by fluid flowing through the first passage **238A** as a result of, e.g., turbulence caused by a velocity of the flowing fluid and a roughness of the surfaces forming the first passage **238A**.

(53) FIG. 7 provides a perspective view of the rotatable heat exchanger **200** as shown in FIG. 3 with the outer casing **202** removed for clarity, according to an exemplary embodiment of the present disclosure. As shown in FIG. 7, the rotatable heat exchanger **200** may be mounted to a shaft(S) such as the L.sub.P shaft **36** or the HP shaft **34**. As shown in FIGS. 5 through 7, each helical passage **238** is further formed or defined by a respective pair of helical turning walls **242**. As shown in FIG. 7, each helical turning wall **242** and each helical passage **238** of the plurality of helical passages **238** extends helically about the annular inner casing **204** (FIG. 3) and/or the axial centerline **216** (FIG. 7) from the first manifold assembly **208** to the second manifold assembly **212** between a respective pair of radially spaced walls or plates **240** (FIGS. 5 and 6 collectively). Each helical passage **238** is fluidly isolated from an adjacent helical passage **238** between the first manifold assembly **208** and the second manifold assembly **212**. In exemplary embodiments, as shown in FIG. 3, the helical passages **238** are all encased between the annular inner casing **204** and

the outer casing **202**.

(54) In various embodiments, each helical passage **238** may make one or more complete turns about the annular inner casing **204** between the first manifold assembly **208** and the second manifold assembly **212**. For example, in one embodiment, as shown in FIG. 7, each helical passage **238** makes one complete turn about the annular inner casing **204** (shown in hidden lines) between the first manifold assembly **208** and the second manifold assembly **212**. In other embodiments, the helical passages **238** make two or more complete turns about the annular inner casing **204** between the first manifold assembly **208** and the second manifold assembly **212**.

(55) In various embodiments, as shown in FIGS. 5 through 7 collectively, the plurality of helical passages **238** includes a plurality of first-fluid passages **338** and a plurality of second-fluid passages **438**. Each first-fluid passage **338** of the plurality of first-fluid passages **338** is fluidly isolated from each second-fluid passage **438**. However, each first-fluid passage **338** is in thermal communication with adjacent (radially and circumferentially) second-fluid passages **438**. As such, thermal energy/heat may be transferred between two fluids flowing respectively through the first-fluid passages **338** and the second-fluid passages **438**.

(56) In exemplary embodiments, as shown in FIGS. 5 and 6, the plurality of helical passages **238** are arranged circumferentially around and radially outwardly from the annular inner casing **204**. The individual first-fluid passages **338** of the plurality of first-fluid passages **338** and the individual second-fluid passages **438** of the plurality of second-fluid passages **438** are arranged or disposed in an alternating pattern both circumferentially and radially about the annular inner casing **204**. In this configuration, each first-fluid passage **338** of the plurality of first-fluid passages **338** is surrounded on at least three sides by three or more second-fluid passages **438** of the plurality of second-fluid passages **438**.

(57) Referring now to FIG. 5, in exemplary embodiments, the first manifold assembly **208** includes a wall or plate **244** which fluidly isolates the first-fluid inlet manifold **218** from the second-fluid outlet manifold **222**. The plate **244** includes or defines a plurality of through-holes or apertures **246**. The first manifold assembly **208** further includes a plurality of conduits or tubes **248**. Each tube **248** of the plurality of tubes **248** is aligned with a respective aperture **246** of the plurality of apertures **246**. In exemplary embodiments, the plurality of tubes **248** are arranged in sets or groups. Each set of tubes **248'** of the plurality of tubes **248** is arranged to align with and provide for fluid flow into a corresponding first-fluid passage **338** of the plurality of first-fluid passages **338**. Each set or group of tubes **248'** extends from the plate **244**, and through the second-fluid outlet manifold **222**.

(58) FIG. 8 provides an enlarged view of a portion of the first end **210** of the core **206** with the outer casing **202** removed and the first manifold assembly **208** attached, according to exemplary embodiments of the present disclosure. As shown in FIGS. 5 and 8 collectively, each respective set of tubes **248'** of the first manifold assembly **208** is fluidly coupled to a corresponding first-fluid passage **338**. As further shown in FIGS. 5 and 8 collectively, the second-fluid outlet manifold (FIG. 5) is open to and in fluid communication with each respective second-fluid passage **438** via respective openings **256** defined in the first manifold assembly **208**. In exemplary embodiments, each opening **256** is circumferentially separated by a respective set of tubes **248'**.

(59) Referring now to FIG. 6, in exemplary embodiments, the second manifold assembly **212** includes a wall or plate **250** which fluidly isolates the first-fluid outlet manifold **226** from the second-fluid inlet manifold **230**. The plate **250** includes or defines a plurality of through-holes or apertures **252**. The second manifold assembly **212** further includes a plurality of conduits or tubes **254**. Each tube **254** of the plurality of tubes **254** is aligned with a respective aperture **252** of the plurality of apertures **252** defined in plate **250**. In exemplary embodiments, the plurality of tubes **254** are arranged circumferentially in sets or groups of tubes **254'**. Each set of tubes **254'** of the plurality of tubes **254** is arranged to align with and provide for fluid flow into a corresponding first-fluid passage **338** of the plurality of first-fluid passages **338**. Each set or group of tubes **254'**

extends from the plate **250**, and through the second-fluid inlet manifold **230**.

(60) FIG. **9** provides an enlarged view of a portion of the second end **214** of the core **206** with the outer casing **202** removed and the second manifold assembly **212** attached, according to exemplary embodiments of the present disclosure. As shown in FIG. **9**, each tube **254** of a respective set of tubes **254'** is fluidly coupled to a corresponding first-fluid passage **338**. As further shown in FIG. **9**, the second-fluid inlet manifold (FIG. **6**) is open to and in fluid communication with each respective second-fluid passage **438** via respective openings **258** defined in the second manifold assembly **212**. In exemplary embodiments, each opening **258** is circumferentially separated by a respective set of tubes **254'**.

(61) FIG. **10** provides a schematic of a bearing cooling system **500** according to an exemplary embodiment of the present disclosure. As shown in FIG. **10**, the bearing cooling system **500** includes the rotatable heat exchanger **200** as illustrated in FIGS. **3-9**, and the bearing compartment **82**, as illustrated in FIG. **2**. In particular embodiments, the bearing cooling system **500** includes a compressor or pump **502**. The pump **502** is suitable to pump a heat transfer fluid/first-fluid (indicated as arrows **504**) such as oil or other suitable coolant from a first-fluid source **506** such as an oil reservoir to the first-fluid inlet **220** of the first-fluid inlet manifold **218** via pipes, conduits, and/or fluid couplings. The first-fluid outlet **228** of the first-fluid outlet manifold **226** is fluidly coupled to an inlet **96** of the bearing cavity **84** of the bearing compartment **82**. An outlet **98** of the bearing cavity **84** is fluidly coupled to the first-fluid source **506**. In particular embodiments, a heat exchanger **508** may be positioned downstream from the outlet **98** of the bearing cavity **84** such that the first fluid **504** passes through the heat exchanger **508** upstream from the first-fluid source **506**. The heat exchanger **508** may be fluidly coupled to a cooling medium source **510**.

(62) In the embodiment shown in FIG. **10**, the compressor section **21** is fluidly coupled to the second-fluid inlet manifold **230** via the second-fluid inlet **232** for providing a second fluid as indicated by arrows **512**, such as the HP compressed air **92**, from the compressor section **21** to the second manifold assembly **212** of the rotatable heat exchanger **200**. The second-fluid inlet **232** may be fluidly coupled to one or more stages of the L.sub.P compressor **22**, the HP compressor **24**, or to one or more stages of both the L.sub.P compressor **22** and the HP compressor **24** via one or more compressor extraction passages or ports **100**.

(63) The second-fluid outlet manifold **222** is fluidly coupled to an inlet **102** of the buffer cavity **86** of the bearing compartment **82** via the second-fluid outlet **224** for providing the second fluid **512** to the buffer cavity **86**. In exemplary embodiments, an outlet **104** of the buffer cavity **86** is fluidly coupled to the turbofan engine **10** at one or more connection points **106** that are downstream from the compressor extraction ports **100** to return the second fluid **512** back to the turbofan engine **10**. For example, the second fluid **512** may be routed back to the turbofan engine **10** at a later compressor stage of the compressor section **21** than the extraction port **100**. In other embodiments, the outlet **104** of the buffer cavity **86** is fluidly coupled to some other portion of the turbofan engine **10** that located fluidly downstream from the compressor section **21** and the extraction ports **100**, such as the combustion section **26**, the HP turbine **28**, the L.sub.P turbine **30**, or to an auxiliary compressor (not shown), auxiliary system, or turbine (not shown).

(64) In operation, the rotatable heat exchanger **200** rotates with the L.sub.P shaft **36** (FIG. **1**). The first fluid **504** flows from the first-fluid source **506** via the pump **502** into the first-fluid inlet manifold **218** via the first-fluid inlet **220**. The first fluid **504** flows through the respective sets of tubes **248'** of the first manifold assembly **208**, and into respective first-fluid passages **338** of the plurality of helical passages **238**. The first fluid **504** flows helically through the respective first-fluid passages **338** about the annular inner casing **204** from the first end **210** of the core **206** to the second end **214** of the core **206**, into respective sets of tubes **254'** of the second manifold assembly **212**, and then into the first-fluid outlet manifold **226**.

(65) As is discussed elsewhere herein, a geometry of the first-fluid passages **338** may influence a size of the pump **502** needed to provide a desired fluid flow through the rotatable heat exchanger

200. For example, when the first-fluid passages **338** are long and narrow, and define a relative high head loss during operation, the pump **502** may need to be larger to overcome the fluidic resistance through the first-fluid passages **338**. By contrast, for embodiments where the passages are short and wide, and define a relatively low head loss during operation, the pump **502** may be smaller while still providing a desired fluid flow.

(66) The first fluid **504** then flows out of the first-fluid outlet manifold **226** via the first-fluid outlet **228** to the inlet **96** of the bearing cavity **84**, thereby providing cooling to one or more bearing assemblies **80** (one bearing assembly shown). The first fluid **504** exits the bearing cavity **84** from the outlet **98** and flows back to the first-fluid source **506**. In particular embodiments, the first fluid **504** may flow through the heat exchanger **508** (when present) to cool the first fluid **504** before it is routed back to the first-fluid source **506**. The first fluid **504** is cooler than the second fluid **512** between the first-fluid source **506** and the bearing cavity **84**.

(67) The second fluid **512** (HP compressed air) flows from one or more of the extraction ports **100** into the second-fluid inlet manifold **230** via the second-fluid inlet **232**. The second fluid **512** flows across the respective sets of tubes **254'** of the second manifold assembly **212** thereby transferring thermal energy/heat from the second fluid **512** to the first fluid **504**. The second-fluid **512** then flows into respective second-fluid passages **438** of the plurality of helical passages **238**. Additional thermal energy/heat is transferred from the second fluid **512** to the first fluid **504** as the second fluid **512** flows helically about the annular inner casing **204** within the respective second-fluid passages **438** from the second end **214** of the core **206** to the first end **210** of the core **206** in a counterflow orientation with respect to the flow direction of the first fluid **504** flowing through the first-fluid passages **338**.

(68) The second fluid **512** then flows into the second-fluid outlet manifold **222** and across the respective sets of tubes **248'** of the first manifold assembly **208** resulting in further heat transfer from the second fluid **512** to the first fluid **504**. The second fluid **512** is then routed out of the second-fluid outlet manifold **222** via the second-fluid outlet **224**, to the buffer cavity **86** to provide the air gap **94** and to pressurize the seal(s) **88** (one shown) between the buffer cavity **86** and the bearing cavity. The second fluid **512** exits the buffer cavity **86** from the outlet **104** and flows back to the turbofan engine **10**. The second fluid **512** may be routed back to the turbofan engine **10** at a later compressor stage of the compressor section **21** than the extraction port **100**. For example, the second fluid **512** may be routed from the outlet **104** of the buffer cavity **86** to one or more of the combustion section **26**, the HP turbine **28**, or the L.sub.P turbine **30** where the second fluid **512** may be used for cooling various engine components or for primary or secondary combustion.

(69) The net result of the thermal interaction between the first fluid **504** and the second fluid **512** as they enter and exit the rotatable heat exchanger **200** is that the temperature of the first fluid **504** increases between the first-fluid inlet **220** and the first-fluid outlet **228** and the temperature of the second fluid **512** decreases between the second-fluid inlet **232** and the second-fluid outlet **224**. The helical shape of the first-fluid passages **338** and the second-fluid passages **438** increases the time provided for heat transfer to occur between the two fluids. In addition, alternating the first-fluid passages **338** and the second-fluid passages **438** circumferentially and radially further increases heat transfer between the two fluids **504**, **512**. Also, heat transfer between the first fluid **504** and second fluid **512** is further enhanced by the centrifugal force acting on both fluids, induced by rotation of the rotatable heat exchanger **200**. The centrifugal force enhances flow turbulence and buoyancy effects, which increases heat transfer.

(70) In other embodiments (not illustrated), the rotatable heat exchanger **200** is configured to enable same flow direction (or co-flow direction) between the first fluid **504** and second fluid **512**. In this configuration the first-fluid inlet manifold **218**, the first-fluid inlet **220**, the second-fluid inlet manifold **230**, and the second-fluid inlet **232** are both defined in one of the first or second manifold assemblies **208**, **212** while the first-fluid outlet manifold **226**, the first-fluid outlet **228**, the second-fluid outlet manifold **222**, and the second-fluid outlet **224** are both defined in the other of

the of the first or second manifold assemblies **208**, **212**.

(71) FIG. **11** provides a cross-sectional schematic view of an alternate embodiment of the rotatable heat exchanger **200** as shown in FIGS. **1-10**, according to the present disclosure. In exemplary embodiments, as shown in FIG. **11**, the first-fluid inlet **220** is defined by a plurality of apertures **260** which provide for fluid communication into the first-fluid inlet manifold **218**.

(72) FIG. **12** provides a schematic of a bearing cooling system **600** according to an exemplary embodiment of the present disclosure. As shown in FIG. **12**, the bearing cooling system **600** includes the rotatable heat exchanger **200** as illustrated in FIG. **11**, and the bearing compartment **82**, as illustrated in FIG. **2**. In this embodiment the apertures **260** of the first-fluid inlet **220** are fluidly coupled to a first-fluid source **606** such as an early-stage extraction port **100(a)** of the L.sub.P compressor **22** (e.g. stages **1** or **2** of the L.sub.P compressor **22**) for providing a first fluid **604** to the first-fluid inlet manifold **218**. The first-fluid outlet **228** fluidly couples the first-fluid outlet manifold **226** back to the L.sub.P compressor **22**.

(73) The second-fluid inlet **232** is fluidly coupled to a later stage extraction port **100(b)** of the L.sub.P compressor **22** to provide a second-fluid **612** to the second-fluid inlet manifold **230**. The second-fluid outlet **224** fluidly couples the second-fluid outlet manifold **222** to the inlet **102** of the buffer cavity **86** of the bearing compartment **82**. The outlet **104** fluidly couples the buffer cavity back to the L.sub.P compressor **22**. In particular embodiments, the bearing cooling system **600** includes a heat exchanger **608** and a cooling medium source **610**. In this configuration, at least a portion of the second-fluid **612** may be routed through the heat exchanger **200** and routed back to the L.sub.P compressor **22**.

(74) In operation, as shown in FIGS. **1-9** and **11-12** collectively, the rotatable heat exchanger **200** rotates with the L.sub.P shaft **36**. The first fluid **604** flows from the first-fluid source **606** into the first-fluid inlet manifold **218** via the first-fluid inlet **220** or apertures **260**. The first fluid **604** flows through the respective sets of tubes **248'** of the first manifold assembly **208**, and into respective first-fluid passages **338** of the plurality of helical passages **238**. The first fluid **604** flows helically through the respective first-fluid passages **338** about the annular inner casing **204** from the first end **210** of the core **206** to the second end **214** of the core **206**, into respective sets of tubes **254'** of the second manifold assembly **212**, and then into the first-fluid outlet manifold **226**. The first fluid **504** then flows out of the first-fluid outlet manifold **226** via the first-fluid outlet **228** and back to the first-fluid source **606** or L.sub.P compressor **22**. The first fluid **604** is cooler than the second fluid **612**.

(75) The second fluid **612** flows from the later stage extraction port **100(b)** of the L.sub.P compressor **22** or from a stage in the HP compressor **24** into the second-fluid inlet manifold **230** via the second-fluid inlet **232**. The second fluid **612** flows across the respective sets of tubes **254'** of the second manifold assembly **212** thereby transferring thermal energy/heat from the second fluid **612** to the first fluid **604**. The second-fluid **612** then flows into respective second-fluid passages **438** of the plurality of helical passages **238**. Additional thermal energy/heat is transferred from the second fluid **612** to the first fluid **604** as the second fluid **612** flows helically about the annular inner casing **204** within the respective second-fluid passages **438** from the second end **214** of the core **206** to the first end **210** of the core **206** in a counterflow orientation with respect to the flow direction of the first fluid **604** flowing through the first-fluid passages **338**.

(76) The second fluid **612** then flows into the second-fluid outlet manifold **222** and across the respective sets of tubes **248'** of the first manifold assembly **208** resulting in further heat transfer from the second fluid **612** to the first fluid **604**. The second fluid **612** is then routed out of the second-fluid outlet manifold **222** via the second-fluid outlet **224**, to the buffer cavity **86** to provide the air gap **94** and to pressurize the seal(s) **88** (one shown) between the buffer cavity **86** and the bearing cavity. The second fluid **612** exits the buffer cavity **86** from the outlet **104** and flows back to the L.sub.P compressor **22**. The second fluid **612** may be routed back to the turbofan engine **10** at a later compressor stage of the compressor section **21** than the extraction port **100(b)**. For

example, the second fluid **612** may be routed from the outlet **104** of the buffer cavity **86** to one or more of the combustion section **26**, the HP turbine **28**, or the L.sub.P turbine **30** where the second fluid **512** may be used for cooling various engine components or for primary or secondary combustion. In an exemplary embodiment, at least a portion of the second fluid **612** may be routed through the heat exchanger **608** for cooling or for further heating before being routed back the L.sub.P compressor **22** or to some other engine system or component.

(77) Again, the net result of the thermal interaction between the first fluid **604** and the second fluid **612** as they enter and exit the rotatable heat exchanger **200** is that the temperature of the first fluid **604** increases between the first-fluid inlet **220** and the first-fluid outlet **228** and the temperature of the second fluid **612** decreases between the second-fluid inlet **232** and the second-fluid outlet **224**. As previously described, the helical shape of the first-fluid passages **338** and the second-fluid passages **438** increases the time provided for heat transfer to occur between the two fluids. In addition, alternating the first-fluid passages **338** and the second-fluid passages circumferentially and radially further increases heat transfer between the two fluids. Also, heat transfer between the first fluid **604** and second fluid **612** is further enhanced by the centrifugal force acting on both fluids, induced by rotation of the rotatable heat exchanger **200**. The centrifugal force enhances flow turbulence and buoyancy effects, which increases heat transfer.

(78) In other embodiment (not illustrated), the rotatable heat exchanger **200** enables same flow direction (or co-flow direction) between the first fluid **604** and second fluid **612**. In this configuration the first-fluid inlet manifold **218**, the first-fluid inlet **220**, the second-fluid inlet manifold **230**, and the second-fluid inlet **232** are both defined in one of the first or second manifold assemblies **208**, **212** while the first-fluid outlet manifold **226**, the first-fluid outlet **228**, the second-fluid outlet manifold **222**, and the second-fluid outlet **224** are both defined in the other of the of the first or second manifold assemblies **208**, **212**.

(79) As alluded to earlier, the inventor discovered, unexpectedly during the course of rotatable heat exchanger and gas turbine engine design—i.e., designing gas turbine engines having a rotatable heat exchanger coupled to a shaft of a turbomachine with various amounts of airflow available to the rotatable heat exchanger, with a variety of geometric ratios of the passages through the heat exchanger, and operable at various rotational speeds, and evaluating an overall engine and heat exchanger performance—a significant relationship exists between the amount of airflow available to the rotatable heat exchanger, a geometric ratio of the passages through the heat exchanger, and a rotational speed of the heat exchanger. The relationship represents an indicator of the ability of a gas turbine engine and rotatable heat exchanger to be able to reject a desired amount of heat from a fluid efficiently when the rotatable heat exchanger is operated at an expected rotational speed, without becoming too difficult to control or being of a design that is too difficult to manufacture.

(80) The relationship applies to a gas turbine engine having a fan and a turbomachine, and defining a bypass ratio of airflow from the fan over the turbomachine to airflow through an inlet of the turbomachine. The relationship ties the amount of airflow available to the rotatable heat exchanger (as represented by the bypass ratio and a radius of the fan) to a geometric ratio of a passage through the heat exchanger and centrifugal force on the fluid when the rotatable heat exchanger, as described in more detail below.

(81) In particular, the inventor discovered that inclusion of a larger fan with a smaller bypass ratio can lead to an increase in airflow available to the rotatable heat exchanger, which may improve an effectiveness of the rotatable heat exchanger. Further, an increase in a length to width ratio of the passages through the heat exchanger, as well as a head loss across each passage, may lead to an increase in heat transfer, creating a more power dense heat exchanger having a greater heat transfer capacity, at least in part due to the increase in surface area for each channel. Further, still, an increase in a centrifugal force acting on the fluid when the rotatable heat exchanger is operating at an expected rotational speed may assist in driving the flow through the heat exchanger.

(82) However, the inventor of the present disclosure found that by bounding the relationship of the

above-noted parameters on an upper end, the rotatable heat exchanger may achieve the above-noted benefits without being prohibitively difficult to control or operate and without being prohibitively difficult to manufacture. In particular, the inventor found that by limiting the length to width ratio of the passages of the heat exchanger, the heat exchanger may be manufactured in a sufficiently durable manner; by limiting the head loss across each channel, the fluid may be provided through the channels using an acceptable amount of energy (e.g., without having to size up one or more pumps for the heat transfer fluid more than desired; see, e.g., pump 502 in FIG. 10); and by limiting the centrifugal force on the fluid and rotatable heat exchanger, and thus the rotational speed of the rotatable heat exchanger, the fluid may flow through the heat exchanger in a controlled manner and the rotatable heat exchanger may be less susceptible to structural issues during operation. The rotatable heat exchanger may be coupled to an L.sub.P shaft to limit the speed in such a manner.

(83) Moreover, the inventor of the present disclosure found that by bounding the relationship of the above-noted parameters on a lower end, the rotatable heat exchanger may further avoid the above-noted deficiencies while exhibiting enough heat rejection capacity to provide a net benefit to the gas turbine engine, despite an increase in weight for the engine and drag on the shaft.

(84) The relationship discovered, infra, can therefore identify an improved engine and rotatable heat exchanger configuration suited for a particular mission requirement, one that takes into account efficiency, weight, thermal capacity needs, complexity, reliability, and other factors influencing the optimal choice for an engine and rotatable heat exchanger configuration.

(85) In addition to yielding an improved gas turbine engine and rotatable heat exchanger, as explained in detail above, utilizing this relationship, the inventor found that the number of suitable or feasible gas turbine engine and rotatable heat exchanger designs capable of meeting the above design requirements could be greatly diminished, thereby facilitating a more rapid down selection of designs to consider as a gas turbine engine and rotatable heat exchanger is being developed. Such a benefit provides more insight to the requirements for a given gas turbine engine and rotatable heat exchanger well before specific technologies, integration and system requirements are developed fully. Such a benefit avoids late-stage redesign.

(86) The desired relationship providing for the improved gas turbine engine, discovered by the inventor, is a Rotating Heat Exchanger Capacity Parameter (RHECP), expressed as:

(87) $\sqrt[4]{\frac{f_{\text{centrifugal}}}{g} \times \frac{f_{\text{channel}}}{H_{\text{channel}}} \times \frac{R_F}{BPR}}$, where f.sub.centrifugal is equal to N.sub.S squared times

R.sub.HX; where N.sub.S is a rotational speed of a shaft to which the rotatable heat exchanger is coupled (in revolutions per minute); where R.sub.HX is a heat exchanger radius of a core of the rotatable heat exchanger; where g is a 9.81 meters per second squared; f.sub.channel is a ratio of L.sub.P to D.sub.P; where L.sub.P is a passage length of a passage of the heat exchanger; where D.sub.P is a passage hydraulic diameter of the passage of the heat exchanger; where $\Delta H_{\text{sub.channel}}$ is a head loss across the first passage when the gas turbine engine is operated at the 100% shaft speed operating condition; where R.sub.F is fan radius; and where BPR is a bypass ratio.

(88) Notably, the inventor of the present disclosure found as a surrogate for the heat exchanger radius (R.sub.HX), the following relationship may be used: a ratio of R.sub.HX to R.sub.S times a ratio of R.sub.S to R.sub.H times a ratio of R.sub.H to R.sub.F times R.sub.F

(R.sub.HX=R.sub.HX/R.sub.S×R.sub.S/R.sub.H×R.sub.H/R.sub.F×R.sub.F). Such a relationship may improve the application of the RHECP to the variety of engine parameters to improve a utility of the RHECP.

(89) Engine parameters and modes of operation dictate rotational speeds, airflows through the engine, and the like at different points of the gas turbine engine. For the above RHECP, the following values are assumed:

(90) TABLE-US-00001 TABLE 1 Ranges appropriate for using Symbol Description Relationships

(1), (2), (3) RHECP Rotating Heat Exchanger 1.8 to 36.1, such as 1.8 to 23.7 Capacity Parameter (dimensionless) f.sub.centrifugal Centrifugal force function 96.5 meters per second squared (m/s.sup.2) to 79,943,796 m/s.sup.2, such as 96.5 m/s.sup.2 to 14,804,407 m/s.sup.2 f.sub.channel Channel size function, equal to 3 to 5,000, such as 3 to 3,000 passage length (L.sub.P) divided by (dimensionless) passage hydraulic diameter (D.sub.P) N.sub.S Shaft rotational speed when the 2,000 revolutions per minute engine is operated at a 100% (rpm) to 30,000 rpm shaft speed operating condition R.sub.HX/R.sub.S Ratio of a heat exchanger radius 1.1 to 5, such as 1.1 to 3 (R.sub.HX) of a core of the rotatable (dimensionless) heat exchanger to a radius of a shaft (R.sub.S) rotatable with the rotatable heat exchanger R.sub.S/R.sub.H Ratio of the radius of the shaft 0.1 to 0.9, such as 0.1 to 0.5 (R.sub.S) to a hub radius (R.sub.H) of the (dimensionless) engine R.sub.H/R.sub.F Ratio of the hub radius (R.sub.H) of 0.1 to 0.9, such as 0.1 to 0.5 the engine to a fan radius (R.sub.F) of (dimensionless) the engine R.sub.F Fan radius of the engine 0.2 meters (m) to 2 m BPR Bypass ratio of the engine 3 to 30 (dimensionless) ΔH .sub.channel Head loss across the first 0.1 m to 200 m, such as 0.1 m to passage when the gas turbine 120 m engine is operated at the 100% shaft speed operating condition

(91) Referring now to FIGS. **13** and **14**, various rotatable heat exchangers in accordance with the present disclosure are provided, such as are encompassed by the values provided above in TABLE 1. FIG. **13** provides a table including numerical values corresponding to several rotatable heat exchangers falling within the ranges depicted in FIG. **14**. FIG. **14** provides a plot **400** of RHECP on a Y-axis **402** and shaft speed on an X-axis **404**.

(92) As shown in FIG. **14**, the plot depicts a first range **406**, with RHECP values greater than or equal to 1.8 and less than or equal to 36.1 and a shaft speed greater than or equal to 2,000 revolutions per minute (rpm) and less than or equal to 30,000 rpm. FIG. **14** additionally depicts a second range **408**, with the RHECP values greater than or equal to 1.8 and less than or equal to 23.7, and the shaft speed greater than or equal to 2,000 rpm and less than or equal to 30,000 rpm.

(93) It will be appreciated that although the discussion above is generally related to a rotatable heat exchanger providing cooling to a bleed airflow from a high pressure compressor and provided to a buffer cavity, in various embodiments of the present disclosure, the relationship outlined above as the RHECP may be applied to any other suitable rotatable heat exchanger configurations.

(94) Further aspects are provided by the subject matter of the following clauses:

(95) A rotatable heat exchanger, comprising: a first manifold assembly; a second manifold assembly; a core extending axially from the first manifold assembly to the second manifold assembly, wherein the first manifold assembly, the core, and the second manifold assembly are circumferentially disposed around and rotate with a shaft, the core comprising: a plurality of helical passages extending from the first manifold assembly to the second manifold assembly, wherein the plurality of helical passages includes a plurality of first-fluid passages fluidly coupled to the first manifold assembly and to the second manifold assembly, and a plurality of second-fluid passages fluidly coupled to the first manifold assembly and the second manifold assembly.

(96) The rotatable heat exchanger of the preceding clause, wherein the first manifold assembly comprises a first-fluid inlet manifold and a second-fluid outlet manifold, and wherein the second manifold assembly comprises a first-fluid outlet manifold and a second-fluid inlet manifold; and wherein the plurality of first-fluid passages is fluidly coupled to the first-fluid inlet manifold and the first-fluid outlet manifold, and wherein the plurality of second-fluid passages is fluidly coupled to the second-fluid outlet manifold and the second-fluid inlet manifold.

(97) The rotatable heat exchanger of any preceding clause, wherein the first manifold assembly comprises a plate positioned to fluidly isolate the first-fluid inlet manifold from the second-fluid outlet manifold; and wherein the first manifold assembly further comprises a plurality of tubes arranged in tube sets, wherein each tube set extends through the second-fluid outlet manifold and fluidly couples the first-fluid inlet manifold to a respective first-fluid passage of the plurality of first-fluid passages.

(98) The rotatable heat exchanger of any preceding clause, wherein the second manifold assembly comprises a plate positioned to fluidly isolate the second-fluid inlet manifold from the first-fluid outlet manifold; and wherein the second manifold assembly further comprises a plurality of tubes arranged in tube sets, wherein each tube set extends through the second-fluid inlet manifold and fluidly couples the first-fluid outlet manifold to a respective first-fluid passage of the plurality of first-fluid passages.

(99) The rotatable heat exchanger of any preceding clause, wherein the second manifold assembly includes a second-fluid inlet in fluid communication with the second-fluid inlet manifold, wherein the second-fluid inlet comprises a plurality of apertures defined along an inner wall of the second manifold assembly.

(100) The rotatable heat exchanger of any preceding clause, wherein the first manifold assembly comprises a first-fluid inlet manifold and a second-fluid inlet manifold, and wherein the second manifold assembly comprises a first-fluid outlet manifold and a second-fluid outlet manifold; wherein the plurality of first-fluid passages is fluidly coupled to the first-fluid inlet manifold and the first-fluid outlet manifold, and wherein the plurality of second-fluid passages is fluidly coupled to the second-fluid inlet manifold and the second-fluid outlet manifold.

(101) The rotatable heat exchanger of any preceding clause, wherein each first-fluid passage of the plurality of first-fluid passages is circumferentially and radially spaced from another first-fluid passage by a respective second-fluid passage of the plurality of second-fluid passages.

(102) The rotatable heat exchanger of any preceding clause, wherein the core comprises an annular inner casing and an outer casing, wherein the plurality of first-fluid passages and the plurality of second-fluid passages are arranged circumferentially about the annular inner casing and radially stacked between the annular inner casing and the outer casing.

(103) A gas turbine engine having a shaft coupled to a compressor section, the gas turbine engine comprising: a bearing compartment including a bearing cavity, a buffer cavity, and a seal disposed between the buffer cavity and the bearing cavity; a first-fluid source providing a first fluid; a second-fluid source providing a second fluid; and a rotatable heat exchanger extending circumferentially around and coupled to the shaft, the rotatable heat exchanger having a core including an annular inner casing and a plurality of helical passages extending helically about the annular inner casing, the plurality of helical passages comprising: a plurality of first-fluid passages fluidly coupled to the first-fluid source, wherein the first fluid flows in a first helical flow direction through the plurality of first-fluid passages; and a plurality of second-fluid passages fluidly coupled to, in serial flow order, the second-fluid source and to the buffer cavity of the bearing compartment, wherein the second fluid flows in a second helical flow direction.

(104) The gas turbine engine of the preceding clause, wherein the first helical flow direction is either counter to the second helical flow direction or same as the second helical flow direction.

(105) The gas turbine engine of any preceding clause, wherein the first-fluid source comprises an oil reservoir or the compressor section of the gas turbine engine; and wherein the second-fluid source comprises the compressor section of the gas turbine engine.

(106) The gas turbine engine of any preceding clause, wherein the rotatable heat exchanger further comprises: a first manifold assembly disposed at a first end of the core, the first manifold assembly defining a first-fluid inlet manifold and a second-fluid outlet manifold; and a second manifold assembly disposed at a second end of the core, the second manifold assembly defining a first-fluid outlet manifold and second-fluid inlet manifold.

(107) The gas turbine engine of any preceding clause, wherein the rotatable heat exchanger further comprises: a first manifold assembly disposed at a first end of the core, the first manifold assembly defining a first-fluid inlet manifold and a second-fluid inlet manifold; and a second manifold assembly disposed at a second end of the core, the second manifold assembly defining a first-fluid outlet manifold and second-fluid outlet manifold.

(108) The gas turbine engine of any preceding clause, wherein the plurality of first-fluid passages

fluidly couples the first-fluid inlet manifold to the first-fluid outlet manifold; and wherein the plurality of second-fluid passages fluidly couples the second-fluid inlet manifold to the second-fluid outlet manifold.

(109) The gas turbine engine of any preceding clause, wherein the first manifold assembly comprises a plate positioned to fluidly isolate the first-fluid inlet manifold from the second-fluid outlet manifold, wherein the first manifold assembly further comprises a plurality of tubes arranged in tube sets, wherein each tube set of the plurality of tubes extends through the second-fluid outlet manifold, and wherein each tube set fluidly couples the first-fluid inlet manifold to a respective first-fluid passage of the plurality of first-fluid passages.

(110) The gas turbine engine of any preceding clause, wherein the second manifold assembly comprises a plate positioned to fluidly isolate the second-fluid inlet manifold from the first-fluid outlet manifold, wherein the second manifold assembly further comprises a plurality of tubes arranged in tube sets, wherein each tube set extends through the second-fluid inlet manifold and fluidly couples the first-fluid outlet manifold to a respective first-fluid passage of the plurality of first-fluid passages.

(111) The gas turbine engine of any preceding clause, wherein the bearing cavity of the bearing compartment is fluidly coupled between the first-fluid outlet manifold and the first-fluid inlet manifold.

(112) The gas turbine engine of any preceding clause, further comprising a heat exchanger in thermal communication with the first fluid between the first-fluid outlet manifold and the first-fluid inlet manifold.

(113) The gas turbine engine of any preceding clause, wherein the buffer cavity of the bearing compartment is fluidly coupled to the second-fluid outlet manifold and wherein the second-fluid inlet manifold is fluidly coupled to the compressor section.

(114) The gas turbine engine of any preceding clause, wherein the core comprises an outer casing, wherein the plurality of first-fluid passages and the plurality of second-fluid passages are arranged circumferentially about the annular inner casing and radially stacked between the annular inner casing and the outer casing.

(115) A bearing cooling system, comprises a rotatable heat exchanger, a bearing compartment including a buffer cavity and a seal, a first-fluid source providing a first fluid. The first fluid source is in fluid communication with the rotatable heat exchanger and the buffer cavity.

(116) A gas turbine engine comprising: a fan section having a fan defining a fan radius ($R_{sub.F}$); a turbomachine comprising a turbine, the gas turbine engine defining a bypass ratio (BPR) of airflow from the fan over the turbomachine to airflow from the fan into an inlet of the turbomachine when the gas turbine engine is operated at a 100% shaft speed operating condition; a shaft coupled to the turbine, the shaft defining a rotational speed ($N_{sub.S}$) when the gas turbine engine is operated at the 100% shaft speed operating condition; a rotatable heat exchanger extending circumferentially around and coupled to the shaft, the rotatable heat exchanger having a core defining a heat exchanger radius ($R_{sub.HX}$) and including an annular inner casing, the rotatable heat exchanger further including a plurality of passages extending about the annular inner casing, the plurality of passages comprising a first passage defining a passage length ($L_{sub.P}$) and a passage hydraulic diameter ($D_{sub.P}$), the rotatable heat exchanger defining a Rotating Heat Exchanger Capacity Parameter (RHECP) greater than or equal to 1.8 and less than or equal to 36.1, RHECP determined according to the following relationship:

(117) $\sqrt[3]{\frac{f_{centrifugal}}{g} \times \frac{f_{channel}}{H_{channel}}} \times \frac{R_F}{BPR}$, where g is a 9.81 meters per second squared; $\Delta H_{sub.channel}$ is a head loss across the first passage when the gas turbine engine is operated at the 100% shaft speed operating condition, with the heat transfer fluid flowing therethrough; $f_{sub.channel}$ is a ratio of $L_{sub.P}$ to $D_{sub.P}$; and $f_{sub.centrifugal}$ is equal to $N_{sub.S}$ squared times $R_{sub.HX}$.

(118) The gas turbine engine of any preceding clause, wherein RHECP is less than or equal to 23.7.

(119) The gas turbine engine of any preceding clause, wherein f.sub.channel is greater than or equal 3 and less than or equal to 5,000.

(120) The gas turbine engine of any preceding clause, wherein f.sub.channel is greater than or equal 3 and less than or equal to 3,000.

(121) The gas turbine engine of any preceding clause, wherein N.sub.S is greater than or equal to 2,000 revolutions per minute (rpm) and less than or equal to 30,000 rpm.

(122) The gas turbine engine of any preceding clause, wherein ΔH .sub.channel is greater than or equal to 0.1 meters and less than or equal to 200 meters.

(123) The gas turbine engine of any preceding clause, wherein the BPR is greater than or equal to 3 and less than or equal to 30.

(124) The gas turbine engine of any preceding clause, wherein the fan section comprises a hub defining a hub radius (R.sub.H), wherein the shaft defines a shaft radius (R.sub.S), wherein R.sub.HX equals a ratio of R.sub.HX to R.sub.S times a ratio of R.sub.S to R.sub.H times a ratio of R.sub.H to R.sub.F times R.sub.F.

(125) The gas turbine engine of any preceding clause, wherein the ratio of R.sub.HX to R.sub.S is greater than or equal to 1.1 and less than or equal to 5, wherein the ratio of R.sub.S to R.sub.H is greater than or equal to 0.1 and less than or equal to 0.9, wherein the ratio of R.sub.H to R.sub.F is greater than or equal to 0.1 and less than or equal to 0.9, and wherein R.sub.F is greater than or equal to 0.2 meters and less than or equal to 2 meters.

(126) The gas turbine engine of any preceding clause, wherein the plurality of passages is a plurality of helical passages.

(127) The gas turbine engine of any preceding clause, wherein the plurality of helical passages comprises a plurality of first-fluid passages fluidly coupled to a first-fluid source, wherein the first fluid flows in a first helical flow direction through the plurality of first-fluid passages; and a plurality of second-fluid passages fluidly coupled to, in serial flow order, a second-fluid source and to a buffer cavity, wherein the second fluid flows in a second helical flow direction.

(128) The gas turbine engine of any preceding clause, wherein the shaft is a low pressure shaft and wherein the turbine is a low pressure turbine.

(129) A method of operating a gas turbine engine, the gas turbine engine having a fan section and a turbomachine and defining a bypass ratio (BPR) over the turbomachine, the method comprising: operating the gas turbine engine at a 100% shaft speed operating condition, wherein operating the gas turbine engine comprises rotating a shaft coupled to a turbine at a rotational speed (N.sub.S) and rotating a rotatable heat exchanger extending circumferentially around and coupled to the shaft to remove heat from a heat transfer fluid, wherein the rotatable heat exchanger includes a core defining a heat exchanger radius (R.sub.HX) and further includes an inner casing, the rotatable heat exchanger further including a plurality of passages extending about the inner casing, the plurality of passages comprising a first passage defining a passage length (L.sub.P) and a passage hydraulic diameter (D.sub.P), the rotatable heat exchanger defining a Rotating Heat Exchanger Capacity Parameter (RHECP) greater than or equal to 1.8 and less than or equal to 36.1, RHECP determined according to the following relationship:

(130) $\sqrt[4]{\frac{f_{\text{centrifugal}}}{g} \times \frac{f_{\text{channel}}}{H_{\text{channel}}} \times \frac{R_F}{BPR}}$, where g is a 9.81 meters per second squared; ΔH .sub.channel is a head loss across the first passage when the gas turbine engine is operated at the 100% shaft speed operating condition, with the heat transfer fluid flowing therethrough; f.sub.channel is a ratio of L.sub.P to D.sub.P; f.sub.centrifugal is equal to N.sub.S squared times R.sub.HX.

(131) The method of any preceding clause, wherein RHECP is less than or equal to 23.7.

(132) The method of any preceding clause, wherein f.sub.channel is greater than or equal 3 and less than or equal to 5,000.

(133) The method of any preceding clause, wherein f.sub.channel is greater than or equal 3 and less than or equal to 3,000.

(134) The method of any preceding clause, wherein N.sub.S is greater than or equal to 2,000 revolutions per minute (rpm) and less than or equal to 30,000 rpm.

(135) The method of any preceding clause, wherein $\Delta H_{\text{sub.channel}}$ is greater than or equal to 0.1 meters and less than or equal to 200 meters.

(136) The method of any preceding clause, wherein the BPR is greater than or equal to 3 and less than or equal to 30.

(137) A rotatable heat exchanger assembly for a gas turbine engine, the gas turbine engine comprising a fan section, a turbomachine, and a shaft, the fan section having a fan defining a fan radius ($R_{\text{sub.F}}$), the gas turbine engine defining a bypass ratio (BPR) of airflow from the fan over the turbomachine to airflow from the fan into an inlet of the turbomachine when the gas turbine engine is operated at a 100% shaft speed operating condition, and the shaft coupled to a turbine of the turbomachine and defining a rotational speed ($N_{\text{sub.S}}$) when the gas turbine engine is operated at the 100% shaft speed operating condition, wherein the rotatable heat exchanger assembly comprises: a rotatable heat exchanger extending circumferentially around and coupled to the shaft when the rotatable heat exchanger is installed in the gas turbine engine, the rotatable heat exchanger having a core defining a heat exchanger radius ($R_{\text{sub.HX}}$) and including an annular inner casing, the rotatable heat exchanger further including a plurality of passages extending about the annular inner casing, the plurality of passages comprising a first passage defining a passage length ($L_{\text{sub.P}}$) and a passage hydraulic diameter ($D_{\text{sub.P}}$), the rotatable heat exchanger defining a Rotating Heat Exchanger Capacity Parameter (RHECP) greater than or equal to 1.8 and less than or equal to 36.1, RHECP determined according to the following relationship:

(138) $\sqrt[4]{\frac{f_{\text{centrifugal}}}{g} \times \frac{f_{\text{channel}}}{H_{\text{channel}}} \times \frac{R_F}{BPR}}$, where g is a 9.81 meters per second squared; $\Delta H_{\text{sub.channel}}$ is a head loss across the first passage when the gas turbine engine is operated at the 100% shaft speed operating condition, with the heat transfer fluid flowing therethrough; $f_{\text{sub.channel}}$ is a ratio of $L_{\text{sub.P}}$ to $D_{\text{sub.P}}$; and $f_{\text{sub.centrifugal}}$ is equal to $N_{\text{sub.S}}$ squared times $R_{\text{sub.HX}}$.

(139) This written description uses examples to disclose the present disclosure, including the best mode and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A gas turbine engine having a shaft coupled to a compressor section and defining a longitudinal axis, a radial direction perpendicular to the longitudinal axis, a circumferential direction around the longitudinal axis, and an axial direction extending parallel to the longitudinal axis, the shaft extends circumferentially about the longitudinal axis, the gas turbine engine comprising: a bearing compartment including a bearing cavity, a buffer cavity, and a seal disposed between the buffer cavity and the bearing cavity; a first-fluid source providing a first fluid; a second-fluid source providing a second fluid; and a rotatable heat exchanger extending circumferentially around and coupled to the shaft, the rotatable heat exchanger having a core including an annular inner casing and a plurality of helical passages extending helically about the annular inner casing, the plurality of helical passages comprising: a plurality of first-fluid passages fluidly coupled to the first-fluid source, wherein the first fluid flows in a first helical flow direction through the plurality of first-fluid passages; and a plurality of second-fluid passages fluidly coupled to, in serial flow order, the second-fluid source and to the buffer cavity of the bearing compartment, wherein the second fluid

flows in a second helical flow direction; wherein the rotatable heat exchanger further comprises: a first manifold assembly disposed at a first end of the core, the first manifold assembly defining a first-fluid inlet manifold having a first-fluid inlet and a second-fluid outlet manifold having a second-fluid outlet, wherein the first-fluid inlet and the second-fluid outlet are each annular and extend around the longitudinal axis and each face in a same axial direction; and a second manifold assembly disposed at a second end of the core opposite the first end of the core, the second manifold assembly defining a first-fluid outlet manifold having a first-fluid outlet and a second-fluid inlet manifold having a second-fluid inlet.

2. The gas turbine engine of claim 1, wherein the first helical flow direction is counter to the second helical flow direction.

3. The gas turbine engine of claim 1, wherein the first-fluid source comprises an oil reservoir of the gas turbine engine; and wherein the second-fluid source comprises the compressor section of the gas turbine engine.

4. The gas turbine engine of claim 1, wherein the first-fluid inlet is radially inward of the second-fluid outlet.

5. The gas turbine engine of claim 4, wherein the plurality of first-fluid passages fluidly couples the first-fluid inlet manifold to the first-fluid outlet manifold; and wherein the plurality of second-fluid passages fluidly couples the second-fluid inlet manifold to the second-fluid outlet manifold.

6. The gas turbine engine of claim 4, wherein the first manifold assembly comprises a plate positioned to fluidly isolate the first-fluid inlet manifold from the second-fluid outlet manifold, wherein the first manifold assembly further comprises a plurality of tubes arranged in tube sets, wherein each tube set of the plurality of tubes extends through the second-fluid outlet manifold, and wherein each tube set fluidly couples the first-fluid inlet manifold to a respective first-fluid passage of the plurality of first-fluid passages.

7. The gas turbine engine of claim 4, wherein the second manifold assembly comprises a plate positioned to fluidly isolate the second-fluid inlet manifold from the first-fluid outlet manifold, wherein the second manifold assembly further comprises a plurality of tubes arranged in tube sets, wherein each tube set extends through the second-fluid inlet manifold and fluidly couples the first-fluid outlet manifold to a respective first-fluid passage of the plurality of first-fluid passages.

8. The gas turbine engine of claim 4, wherein the bearing cavity of the bearing compartment is fluidly coupled between the first-fluid outlet manifold and the first-fluid inlet manifold.

9. The gas turbine engine of claim 4, further comprising an additional heat exchanger in thermal communication with the first fluid between the first-fluid outlet manifold and the first-fluid inlet manifold.

10. The gas turbine engine of claim 9, wherein the first fluid travels in a first-fluid flowpath from the first-fluid source to the first-fluid inlet via a pump, through the plurality of first-fluid passages of the rotatable heat exchanger in the first helical flow direction, out the first-fluid outlet, to a bearing cavity inlet of the bearing compartment, through the bearing cavity to provide cooling to a bearing assembly contained in the bearing cavity, out a bearing cavity outlet, to the additional heat exchanger that cools the first fluid before traveling back to the first-fluid source.

11. The gas turbine engine of claim 10, wherein the second fluid travels in a second-fluid flowpath from a second-fluid source outlet of the second-fluid source to the second-fluid inlet, through the plurality of second-fluid passages of the rotatable heat exchanger in the second helical flow direction, out the second-fluid outlet, to a buffer cavity inlet of the bearing compartment, through the buffer cavity to provide an air gap and to pressurize the seal between the buffer cavity and the bearing cavity, out a buffer cavity outlet, and back to the second-fluid source via a second-fluid source inlet.

12. The gas turbine engine of claim 11, wherein the second-fluid source outlet is in fluid communication with the compressor section of the gas turbine engine, and wherein the second fluid is a high pressure compressed air from the compressor section.

13. The gas turbine engine of claim 12, wherein the second-fluid source inlet is in fluid communication with the compressor section and a turbine section of the gas turbine engine.
 14. The gas turbine engine of claim 4, wherein the buffer cavity of the bearing compartment is fluidly coupled to the second-fluid outlet manifold and wherein the second-fluid inlet manifold is fluidly coupled to the compressor section.
 15. The gas turbine engine of claim 4, wherein the core comprises an outer casing, wherein the plurality of first-fluid passages and the plurality of second-fluid passages are arranged circumferentially about the annular inner casing and radially stacked between the annular inner casing and the outer casing.
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